

Talk to Krishna — Mobile App Frontend Integration Guide

This guide details what needs to be implemented by the **Mobile App Development Team** (iOS & Android) to support partner-attribution and campaign tracking.

🎯 1. Objective: What are we tracking?

When a user scans a partner's QR code (e.g. from a printed banner or temple standee) and installs "Talk to Krishna", the mobile app must detect **which organization** and **which campaign** generated this installation.

This enables the system to credit the correct temple partner for downstream user sign-ups and donations.

🔍 2. Why is this necessary?

App Stores do not automatically share referral links with installed apps.

- On **Android**, Google Play supports a metadata transmission API (Install Referrer).
- On **iOS**, App Store blocks this, so we use probabilistic fingerprinting.

Thus, the frontend app must collect install parameters at the first app startup and forward them to your signup endpoint.

📱 3. Android Frontend Integration (Google Play Referrer)

Android apps must query the **Google Play Install Referrer API** immediately upon first launch to check if a referral token was stored during installation.

Implementation Steps:

- Add the standard dependency to your App's build.gradle file:

```
implementation 'com.android.installreferrer:installreferrer:2.2'
```

- Initialize the client and fetch the referrer parameters:

```
import com.android.installreferrer.api.InstallReferrerClient
import com.android.installreferrer.api.InstallReferrerStateListener

val referrerClient = InstallReferrerClient.newBuilder(context).build()
referrerClient.startConnection(object : InstallReferrerStateListener {
    override fun onInstallReferrerSetupFinished(responseCode: Int) {
        if (responseCode == InstallReferrerClient.InstallReferrerResponse.OK) {
            val response = referrerClient.installReferrer
            val referrerUrl = response.installReferrer
            // Example string returned: "org_id=org-iskcon-delhi&campaign_id=camp-del-01"

            // Parse keys from URL format
            val params = parseReferrerUrl(referrerUrl)
            val orgId = params["org_id"]
            val campaignId = params["campaign_id"]

            // Save locally (SharedPreferences) to pass to your signup API
            saveAttributionLocally(orgId, campaignId)
        }
    }
    override fun onInstallReferrerServiceDisconnected() {
        // Retry connection logic if required
    }
})
```

🍏 4. iOS Frontend Integration (Device Fingerprint Profiling)

Apple blocks install referrers. To bypass this, the iOS App collects basic device specifications at first launch. This profile is compared in the backend against recent web QR scan events.

Implementation Steps:

- Capture these 5 device characteristics at the first launch:
 - Public IP Address:** Fetch the user's current public IP address (using a lightweight JSON query e.g. <https://api.ipify.org?format=json>).
 - Device Model:** The hardware name of the iPhone (e.g. iPhone 15 Pro).
 - OS Version:** The installed iOS version (e.g. 17.4).
 - Screen Resolution:** Logical screen width and height (e.g. 1179x2556).
 - Timezone:** User's local timezone settings (e.g. Asia/Kolkata).
- Store this profile in a dictionary structure:

```
struct DeviceProfile: Codable {
    let ipAddress: String // e.g. "122.161.49.52"
    let deviceModel: String // e.g. "iPhone15,2"
    let osVersion: String // e.g. "17.4"
    let screenResolution: String // e.g. "1179x2556"
    let timezone: String // e.g. "Asia/Kolkata"
}
```

🌐 5. Sign-Up API Interface Specification

When a user registers or performs onboarding, the app frontend must send the collected attribution data to your API endpoint.

Request Payload Example:

The app should include an attribution_data object in the sign-up body:

```
{
  "username": "bhakta_raj",
  "email": "user@example.com",
  "device_platform": "iOS", // "iOS" or "Android"
  "attribution_data": {
    // For iOS (device profile specs)
    "ip_address": "122.161.49.52",
    "device_model": "iPhone 15 Pro",
    "os_version": "17.4",
    "screen_resolution": "1179x2556",
    "timezone": "Asia/Kolkata",

    // For Android (Direct parsed URL values)
    "android_referrer_org": "org-iskcon-delhi",
    "android_referrer_campaign": "camp-del-01"
  }
}
```

[TIP]

Performance Recommendation: Always query the referrer/IP asynchronously in the background. Do not block the launch screen or core onboarding steps if the network takes time to retrieve the public IP address.